

A Mostly Harmless Introduction to Claude Code, Codex and Friends

For Empirical Researchers

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Welcome

- Workshops for Ukraine
- Audience: empirical researchers, not software engineers
- Scope: practical and hands-on

About Me

Daniel Sánchez, MA

- Economist, MA in Economics, Simon Fraser University (Canada)
- Statistician, Government of Alberta: nowcasting economic accounts in R/SQL
- Co-director, Laboratorio de Investigación en Desarrollo del Ecuador (LIDE), an Ecuadorian development research lab

Working with agentic AI

- Design GenAI workflows (Claude, ChatGPT, Gemini, Copilot) for due diligence, financial analysis, and reporting
- Build autonomous agents with Claude Code and Cowork for document processing and structured extraction
- Connect agents to everyday tools (Excel, Outlook, Teams) via MCP

Agenda

- 1 Introduction: why agentic AI
- 2 What's out there: the landscape of tools
- 3 Cybersecurity: do's and don'ts
- 4 How to use agentic AI: mechanics and cost models
- 5 Quick overview: best model for common users
- 6 Usages (the core, hands-on section)
- 7 Multi-agent (tentative)

Introduction: Why Agentic AI

The real bottleneck in most empirical projects is not the econometrics. It's the grind around it:

- Cleaning a messy microdata file for the third time this month
- Chasing down why a regression coefficient changed after “nothing changed”
- Reformatting the same table for a fourth journal's style

That grind is exactly the part a chat window cannot touch, and exactly the part an agent can sit inside.

Vanilla AI vs. Agentic AI

Vanilla AI (ChatGPT, Claude.ai, a chat window)

- You paste in code or data, it replies with text
- You copy the suggestion back out, apply it yourself, rerun it yourself
- It never sees whether its own suggestion actually worked

Agentic AI (Claude Code, Codex, and friends)

- It opens your files, makes the edit itself, runs the script itself
- It reads the real error or output and tries again, without you relaying it
- The unit of delegation moves from *a question* to *a task*

From Autocomplete to Agents: What Changed

- Autocomplete and chat: one-shot generation, no feedback loop, no way to check its own answer against reality
- Agents run a loop instead: **read** → **plan** → **act** → **observe** → **repeat**
 - “Act” means calling a real tool: open a file, run a script, query data
 - “Observe” means reading the actual result, not guessing at it
- What made this possible: larger context windows and reliable tool use, not simply “a smarter model”

Why This Matters for Empirical Research

- The agent can read your raw data, write and run the cleaning script, and notice when a value label silently changed across survey waves
- It can run your regression, read the actual error message, and fix the bug, instead of you pasting the traceback back and forth
- It does **not** replace your judgment on identification, specification, or interpretation: you still decide what's correct, it does the retyping and rerunning
- Net effect: fewer hours lost to mechanical iteration, more time on the parts of the research that actually require you

The Landscape: What's Out There

- Terminal / CLI agents: Claude Code, Codex, and friends
- Chat-based assistants vs. agentic tools
- IDE-integrated agents
- What researchers are actually adopting right now

Cybersecurity: Do's

- Use vetted, org-approved tools and check for a data processing agreement
- Keep credentials out of prompts: use secret managers/env vars, never paste tokens or passwords into chat
- Scope MCP/data connections to least privilege (read-only where possible)
- Know what data leaves your machine before granting file or network access

Cybersecurity: Don'ts

- Don't paste PII, credentials, or restricted admin microdata into a public chat interface
- Don't grant blanket filesystem or internet access without understanding what the agent can reach
- Don't treat "the agent says it's fine" as a compliance answer
- Don't skip your institution's confidentiality and data-handling agreements for convenience

Mechanics and Cost Models

- How an agentic session actually works under the hood
- Context windows, tool calls, and iteration loops
- Subscription vs. API/token-based pricing
- Ballpark costs for a typical research workflow

Quick Overview: Best Model for Common Users

- What “best” depends on: task type, budget, workflow
- Rules of thumb for picking a model/tool as a non-specialist
- When the default choice is good enough

Usages

The core, hands-on section.

Usages: Code Review

- Catching bugs and silent errors in statistical/data code
- Reviewing before merging or before submitting to a coauthor

Usages: Literature Review

- Search, summarize, and synthesize across papers
- Where agents help, and where they still hallucinate citations

Usages: Writing Review

- Drafting and editing papers, referee reports, slides
- Keeping the researcher's voice and claims intact

Usages: Skills and Agents from Modern Economists

- Reusable prompts, skills, and workflows built by other researchers
- Examples worth adapting

Usages: MCP for Data Access

- Connecting the agent directly to data sources instead of manual CSV exports
- Databases, Google Drive, Notion, internal government systems, APIs
- Relevant to this audience: StatCan, IESS/REESS, ENEMDU-style admin data pulls

Multi-Agent

(Tentative, first thing to cut if running long)

- Running multiple agents in parallel or in a pipeline
- When one agent isn't enough: division of labor, orchestration
- Practical caveats: cost, coordination, verification overhead

Thank You

- Daniel Sánchez Pazmiño
- Workshops for Ukraine, September 3, 2026